# SEMPLIFICAZIONE DI FRAZIONI ALGEBRICHE

Es pag. 651 n. 455

$$\frac{2x^2+5x+2}{x^2+7x+10} = \frac{\cancel{(x+2)}(2x+1)}{\cancel{(x+2)}(x+5)} = \frac{2x+1}{x+5}$$

1) $2x^2+5x+2=0$ $\quad \Delta = b^2-4ac = (5)^2-4\cdot 2\cdot 2 =$

$= 25-16 = 9$

equazione omogenea

$$x_{1/2} = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-5 \pm \sqrt{9}}{2 \cdot 2} = \frac{-5 \pm 3}{4} = \begin{cases} \frac{-5-3}{4} = \frac{-8}{4} = -2 \\ \frac{-5+3}{4} = -\frac{2}{4} = -\frac{1}{2} \end{cases}$$

$x_1 = -2$ ; $x_2 = -\frac{1}{2}$

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

$$2x^2 + 5x + 2 = 2(x+2)\left(x + \frac{1}{2}\right) = \cancel{2}(x+2)\left(\frac{2x+1}{\cancel{2}}\right) =$$

$$= (x+2)(2x+1)$$

2) $x^2 + 7x + 10 = 0$ → equazione associata

$$\Delta = b^2 - 4ac = (7)^2 - 4 \cdot 1 \cdot 10 = 49 - 40 = 9$$

$$x_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-7 \pm \sqrt{9}}{2} = \frac{-7 \pm 3}{2} = \begin{cases} \frac{-7-3}{2} = -\frac{10}{2} = -5 \\ \frac{-7+3}{2} = -\frac{4}{2} = -2 \end{cases}$$

$x_1 = -5$, $x_2 = -2$

$$x^2 + 7x + 10 = 1 \cdot (x+5)(x+2)$$

Es. pag. 651 n. 450

$$\frac{6x^2-7x+2}{4x^2-2x} = \frac{\cancel{(2x-1)}(3x-2)}{2x\cancel{(2x-1)}} = \frac{3x-2}{2x}$$

1) $6x^2-7x+2=0$ → equazione monomia

$$\Delta = b^2-4ac = (-7)^2-4\cdot 6\cdot 2 = 49-48 = 1$$

$$x_{1/2} = \frac{-b\pm\sqrt{\Delta}}{2a} = \frac{7\pm\sqrt{1}}{2\cdot 6} = \frac{7\pm 1}{12} = \begin{cases} \frac{7-1}{12} = \frac{6}{12} = \frac{1}{2} \\ \frac{7+1}{12} = \frac{8}{12} = \frac{2}{3} \end{cases}$$

$$6x^2-7x+2 = 6\left(x-\frac{1}{2}\right)\left(x-\frac{2}{3}\right) = \cancel{6}\left(\frac{2x-1}{\cancel{2}}\right)\left(\frac{3x-2}{\cancel{3}}\right) = (2x-1)(3x-2)$$

d) $4x^2 - 2x = 0$ → equazione spaccata

$2x^2 - x = 0$ → $x(2x-1) = 0$ $\begin{cases} x = 0 \\ 2x - 1 = 0 \end{cases}$

$2x = 1 \rightarrow x = \frac{1}{2}$

$x_1 = 0$ , $x_2 = \frac{1}{2}$

$4x^2 - 2x = 4(x-0)\left(x - \frac{1}{2}\right) = \cancel{4}^{2} \cdot x \left(\frac{2x-1}{\cancel{2}_{1}}\right) = 2x(2x-1)$